LESSON 2.2

Represent and Describe Data in a Histogram

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can make and read a histogram to display data in intervals.

WHY THIS MATTERS The basketball league has 30 players and wants to display everyone's points-per-game average in a way that shows how the data is distributed. The raw averages are: 2, 5, 7, 8, 9, 10, 11, 11, 12, 13, 13, 14, 14, 15, 15, 16, 16, 17, 18, 18, 19, 20, 21, 22, 23, 24, 25, 28, 30, 32. Listing all 30 numbers is hard to read — a histogram will reveal the pattern!

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Histogram *Spanish: Histograma*	A bar graph that groups data into equal ranges. The bars touch.	Bars that touch
Frequency *Spanish: Frecuencia*	How many times a value shows up.	Bar height shows the count
Interval *Spanish: Intervalo*	A range of numbers used to group data.	10-19 is one interval
Distribution *Spanish: Distribución*	How the data is spread out.	Tallest bar in the middle
Variability *Spanish: Variabilidad*	How spread out the numbers are.	Data in just 2 intervals = low variability. Data across 6 intervals = high variability
Frequency table *Spanish: Tabla de frecuencias*	A table that lists each interval and how many values fall in it.	10-19 . . . 3

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Walk times (min): 4, 15, 9, 22, 11, 18, 27. Use 0–9, 10–19, 20–29.

1. 0–9 holds 4 and 9: frequency 2.
2. 10–19 holds 15, 11, 18: frequency 3. 20–29 holds 22, 27: frequency 2.
3. Draw touching bars of height 2, 3, 2.

Answer: Check: 2 + 3 + 2 = 7 values

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

New data: 2, 4, 8, 11, 13, 16, 21. We will use intervals of 10 again: 0–9, 10–19, 20–29.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- My intervals are _____ wide and they do not _____ .
- The interval _____ holds _____ values.
- The tallest bar is _____ because _____ .
- The bars touch because the intervals _____ .

Word bank histogram · interval · frequency · tally · equal · overlap · bar · height · count · touch · distribution · data

WATCH OUT!

A common mistake

A common mistake in Display Data: Histograms is counting a boundary value in two intervals at once

LESSON 2.2

Represent and Describe Data in a Histogram

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

In a histogram, what does the height of each bar represent?

(A) The frequency (count) of data in that interval

(B) The average of the data in that interval

(C) The range of the interval

(D) The median of the data

STEPS TO TRY

1. Think about what you'd count to draw one bar: you'd tally how many data values land inside that interval.
2. Height is a COUNT, not a calculation — no averaging, no middle values. Ask which choice simply says 'how many.'
3. First step: imagine the interval 10–14 holding the values 11, 12, 12, 14. Its bar would rise to 4 — because 4 values fall inside.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A histogram shows these frequencies: 0–4: 3 players, 5–9: 8 players, 10–14: 12 players, 15–19: 5 players. How many players are represented in total?

(A) 12

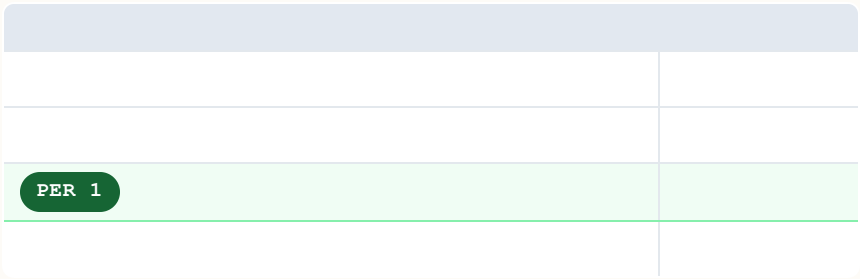
(B) 15

(C) 20

(D) 28

STEPS TO TRY

1. The question asks for the TOTAL across all four intervals — not the tallest bar and not the biggest interval.
2. Each frequency counts the players in one interval, and no player is counted twice — so combine all four frequencies into one total.
3. Set up the addition: 3 + 8 + 12 + 5. The sum of all the bar heights is the total number of players.



1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

3 **Circle the letter of the best answer. Show how you know.**

How is a histogram different from a regular bar graph?

(A) Histogram bars touch (no gaps) because the intervals are continuous ranges

(B) There is no difference

(C) Histograms use colors and bar graphs don't

(D) Bar graphs show numbers and histograms show words

STEPS TO TRY

1. Picture both graphs side by side: one has spaces between its bars, one doesn't. Ask why that would be.
2. Think about the x-axis: categories like 'red, blue, green' can be separated, but number intervals like 0–9 and 10–19 connect end to end.
3. First step: write two neighboring intervals, 0–9 and 10–19. Notice that 10 starts right where 9 ends — that's why the bars behave differently.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A histogram has intervals: 0–4, 5–9, 10–14, 15–19. What is the width of each interval?

(A) 4
(B) 5
(C) 10
(D) 19

STEPS TO TRY

1. Width means how many whole-number values fit inside one interval — count them, don't just subtract the endpoints.
2. List the values in the first interval, 0–4: count every whole number from 0 up to and including 4.
3. Count aloud: 0, 1, 2, 3, 4 — how many numbers did you say? Check that the interval 5–9 holds the same amount.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

The tallest bar is the 10–19 interval with 16 players. What does the shape of this histogram tell you about scoring in the league?

SENTENCE STARTER Most players score between ___ and ___ points per game. The histogram is skewed ___ because ___. This tells me that ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help